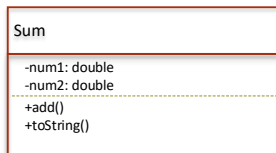

From class diagram to program

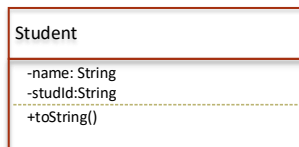
Note: Complete the exercises below before the next lecture

Exercise 1: The diagram below shows the class **Sum**. The method **add()** returns the sum of the two numbers (**num1** and **num2**); **toString()** returns the numbers and their sum as a string



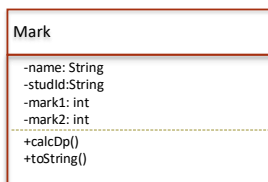
- Write the class in Java; include:
 - At least two constructors and
 - Get and set methods for each data member
- Write a “tester” or “driver” (**TestSum**), which contains **main()** and “uses” class **Sum**. This program must display at least **two** instances of **Sum**.

Exercise 2: The diagram below shows class **Student**. The **toString()** method returns **name** and student Id (**studId**) as string.



- Write the class in Java; include:
 - At least two constructors and
 - Get and set methods for each data member
- Write a program (**TestStudent**), “tester” or “driver” which “uses” the class **Student**. This program must create and display at least **two** instances of **Student**.

Exercise 3: The diagram below shows class **Mark**. The method **calcDp()** returns the dp (i.e., the average of the test marks) of a student; **toString()** returns the data members and the dp as a string.



- Convert the class **Mark** to a Java program.
- Write a “tester” program (**Tester**) which “uses” the class **Mark**. This program must create and display at least two instances of **Mark**.

Exercise 4: The diagram below shows class **Item**. **totalPrice()** returns total price (i.e., the price and VAT chargeable on the item). VAT is calculated as 15% of the price. **toString()** returns the data members and the total price as a string.

Item
-code: String -name: String -price: double
+totalPrice(): double +toString(): String

1. Convert the class **Item** to a Java program.
2. Write a “tester” program (**TestItem**) which “uses” the class **Item**. This program must create and display at least two instances of **Item**.